JENSEN PRODUCTION

Fleet Management System

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Design Review Document

A walkthrough of how the system understands your business

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Schema version: 1.1

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How to use this document

This document is your chance to confirm — or correct — how the new Fleet Management System has been designed to work. Before any code gets written, we want to walk you through the system's understanding of your business so you can push back on anything that doesn't match how Jensen Production actually operates.

The design has been built from four sources: the whiteboard sketches you produced, the parts spreadsheet you shared, the prototype dashboard you started in Gemini, and the conversations we've had since.

What this document is — and isn't

What it IS:

- A plain-English walkthrough of how the system thinks about parts, customers, bikes, orders, and maintenance.
- A list of decisions that were made on your behalf — with specific questions where we need your input.
- A scenario walkthrough that shows what happens, end to end, when an order comes in.

What it is NOT:

- A technical specification. There are no database tables or code in this document.
- A finished design. Nothing here is locked in until you've signed off.
- Comprehensive. We've focused on the things most likely to need correction.

How to give feedback

You can do any of the following — whatever works best for you:

1. Open this document in Word and use Track Changes or Comments to mark things up directly.
2. Print it out, write on it with a pen, and scan or photograph the marked-up pages.
3. Make a list of corrections in a separate email or call.

Wherever you see a yellow box marked QUESTIONS FOR DENNIS, that's where we genuinely need your input. There are about 30 such questions in this document, organized by topic.

Total time to read and respond: roughly 30–45 minutes. You don't have to do it in one sitting — the document is structured so you can stop and start.

The big picture

The Fleet Management System tracks six things, all of them connected:

What	What it covers
Parts	Every component you buy or stock — frames, motors, cables, bolts, accessories.
Suppliers	Companies you buy parts from. Today: Eastek HK. Tomorrow: as many as you need.
Customers	Hotels, hospitals, municipalities, FM companies, businesses, individuals — anywhere in the world.
Bikes	Every physical bike you've built or are building, with full lifecycle history.
Orders	Sales orders from customers, plus internal manufacturing orders for mechanics.
Maintenance	Repair tickets, work orders, parts consumed in service, and service agreements.

These six are connected like this: parts come in from suppliers; bikes get built from parts; bikes are sold or assigned to customers; customers report problems; service agreements determine who pays for repairs.

Everything is tracked over time. The system can answer questions like:

- "What was our cost basis on Bafang motors purchased in 2024?"
- "Which bikes did we build for Københavns Kommune last year?"
- "What's the maintenance history of bike JP-2026-HSB-001?"
- "How many e-bikes are currently in service vs. needing repair?"
- "What's the total value of our parts inventory in DKK right now?"

A walkthrough: “Aarhus Universitetshospital orders 50 service bikes”

This is the workflow we’ve designed for a typical custom B2B order. Read through it and tell us where it diverges from how things actually happen at Jensen Production.

Step 1 — The customer

Aarhus Universitetshospital calls. They want 50 service bikes for hospital-wide use. The system has them on file as a customer with their CVR number, EAN number for public-sector e-invoicing, billing in DKK, segment "Hospital", and a "Corporate" pricing tier.

Step 2 — An offer goes out

You build an offer for 50 × Hospital Service Bike (53cm, white). The offer references one of your existing bike templates — a saved recipe of parts that defines what goes into this bike model. The offer has a price, an expiry date, and is sent to the hospital’s procurement contact.

Step 3 — The offer is accepted

The offer becomes a sales order. The sales order is the commercial commitment: who, what, when, how much.

Step 4 — A manufacturing order is created

Internally, the sales order spawns a manufacturing order. This is the build directive for your shop: "build 50 bikes against template X." The manufacturing order has a target quantity (50), planned start and finish dates, and gets assigned to whoever is leading that build.

A sales order and a manufacturing order are kept separate on purpose. The sales order is your agreement with the customer; the manufacturing order is your internal production plan. One sales order can spawn multiple manufacturing orders if needed (e.g., "20 bikes for delivery in June, 30 for delivery in August").

Step 5 — Template loaded, minor adjustments made

The template’s parts list — perhaps 47 components, from frame down to reflectors — is loaded into the dashboard. You or the lead mechanic review it and make small adjustments specific to this order. For instance, the customer requested upgraded display units, so you substitute one display part for another. The system records what was changed and why.

Step 6 — Mechanics build

Each bike is built and gets assigned its identifiers as it’s completed. For an electric bike, that means all five from your whiteboard:

- Frame number
- Lock number

- Battery lock number
- Battery number
- Charger number

For pedal bikes, only frame and lock are required. The system enforces this per bike type — it won't ask you for a charger number on a pedal bike.

As each bike is completed, its status changes from "building" to "in_stock", and the manufacturing order's "completed" counter ticks up: 1 of 50, 2 of 50, and so on.

Step 7 — Bikes go to the customer

When all 50 are built, the bikes are delivered. Each bike's status changes from "in_stock" to "assigned" with Aarhus Universitetshospital recorded as the owner. The manufacturing order is marked "completed."

Step 8 — Invoice

An invoice is generated against the sales order. Since this is a Danish hospital, it's a DKK invoice with 25% Danish moms and includes the EAN number for public-sector e-invoicing. The invoice gets a unique number (e.g., INV-2026-0042) and can be exported to e-conomic.

Step 9 — Service kicks in

If Aarhus has a service agreement with you, future repairs on these bikes are tracked but billing depends on what the agreement covers. Maintenance tickets, work orders, and parts consumption are all logged against each bike's frame number, building up a complete service history.

QUESTIONS FOR DENNIS — The build workflow

Q1. Is this the workflow as you actually run it? What's missing or wrong?

Q2. Do you ever skip steps — e.g., go straight to a sales order without sending a formal offer first?

Q3. Do you ever build bikes WITHOUT a customer order, just to keep stock on hand?

Q4. Are templates reused often, or do most orders need significant customization?

Q5. When you make adjustments to a template for an order, what's typical: substituting parts, adding extras, removing optional parts, or all of these?

Q6. Who in the shop typically does each step? (You? A specific mechanic? Multiple people?)

Bike types

The system distinguishes between different kinds of bikes because they have different requirements (e.g., pedal bikes don't have batteries; tricycles might have different identifiers). Here are the bike types we've set up:

Type	Danish	Description
Pedal Bike	Pedalcykel	Standard human-powered bike, no electric assist
Electric Bike	Elcykel	Electric pedal-assist with motor and battery
Tricycle	Trehjulet cykel	Three-wheeled (pedal or electric)
Cargo Bike	Ladcykel	Cargo bike for goods or passengers
Hand Cycle	Håndcykel	Hand-powered, accessibility use
Child Bike	Børnecykel	Sized for children
Other	Andet	Anything not covered above

QUESTIONS FOR DENNIS — Bike types

Q1. Are these all relevant to what you build, or are some of these we don't need?

Q2. What's missing? (Recumbent, tandem, folding, racing, MTB?)

Q3. Do you make a meaningful distinction between pedal cargo and electric cargo, or is "Cargo" enough as one category?

The five bike identifiers

Your whiteboard called out five unique numbers per bike: frame number, lock number, battery lock number, battery number, charger number. These are now formally part of the system, but with a twist — not every bike type needs all five.

Here's how we've set up which identifiers each bike type requires:

Bike Type	Required identifiers
Pedal Bike	Frame + Lock
Electric Bike	Frame + Lock + Battery Lock + Battery + Charger (all 5)
Tricycle	Frame + Lock
Cargo Bike	Frame + Lock (battery items optional if electric variant)
Hand Cycle	Frame
Child Bike	Frame
Other	Frame

The system also supports additional identifier types that aren't required but can be recorded if relevant: QR code, RFID tag, Apple AirTag UUID, GPS tracker IMEI, SIM card number, Bluetooth ID, motor serial, display serial. Adding a new identifier type in the future (say, a future Smart Lock standard) is a one-line change — we don't have to redesign anything.

QUESTIONS FOR DENNIS — Identifiers

Q1. For an e-bike, are these five identifiers correct and complete? Anything missing?

Q2. For a pedal bike, is "Frame + Lock" right, or do you need anything else captured?

Q3. Do you ever attach more than one lock to a bike (e.g., a frame lock plus a chain lock)?

Q4. Do any of your bikes today have GPS trackers, AirTags, or RFID tags installed?

Q5. Should "Charger Number" be required on every e-bike, or do you sometimes deliver without a dedicated charger (e.g., shared chargers)?

Customer segments

Customers are categorized by what kind of organization they are. This drives reporting (e.g., "show me all our hospital customers"), default behaviors, and sometimes pricing. Here's the list:

Segment	Danish	Examples
Hotel	Hotel	Hotels and hospitality — your core segment
Hospital	Hospital	Hospitals and healthcare facilities
Municipality	Kommune	Danish kommuner, regions
Facility Management	Facility Management	FM companies managing buildings/sites
Business (B2B)	Erhverv (B2B)	Generic business customers
Consumer (B2C)	Privat (B2C)	Private individuals
Rental Company	Udlejningsfirma	Bike rental / sharing operators
Other	Andet	Anything not covered above

QUESTIONS FOR DENNIS — Customer segments

Q1. Are these the right segments for the customers you actually have today?

Q2. Is "Hotel" really a separate segment from B2B in your mind, or is it just a label?

Q3. Are there segments missing? (Schools? Universities? Government agencies? Embassies? Tour operators?)

Q4. Do you sell to international customers today, and if so what countries?

Customer pricing groups

Separate from segments, the system has pricing groups that determine default discounts. The prototype dashboard you started in Gemini suggested three:

Pricing Group	Default Discount	Used for
Retail	0%	Walk-in or one-off customers, full retail price
Wholesale	15%	Wholesale partners, distributors
Corporate	20%	Large corporate accounts, public sector

A customer is assigned to one pricing group; their default discount applies to their orders unless overridden line by line.

QUESTIONS FOR DENNIS — Pricing

Q1. Are 0% / 15% / 20% the right discount percentages, or are they just placeholder numbers?

Q2. Do you have other tiers in practice (e.g., a "Strategic" tier with bigger discounts for repeat hospital customers)?

Q3. Do you negotiate per-customer discounts that override the group default?

Q4. Do you ever apply different discounts on different lines of the same order (e.g., 20% off bikes but 10% off accessories)?

Q5. Are there ever volume discounts — the more they buy, the bigger the discount?

The bike lifecycle

Every bike in the system moves through a defined lifecycle. Each transition is logged automatically with a timestamp, so you always have the full history of where a bike has been. Here are the states:

State	Means	Typical next state
Planning	Order placed, build not started	Building
Building	Mechanics actively building	In Stock
In Stock	Built, frame number assigned, no customer yet	Assigned
Assigned	Sold/assigned to a customer	In Service
In Service	Actively in use by customer	In Maintenance
In Maintenance	Pulled in for repair	In Service
Retired	End of life, not in use	—
Lost or Stolen	Reported missing	—

QUESTIONS FOR DENNIS — Lifecycle

Q1. Does this match how you think about a bike's life from build to retirement?

Q2. Is "Assigned" different from "In Service" in any meaningful way for you, or could those be one state?

Q3. Is there a stage we're missing? (Quality check before stock? Awaiting parts during maintenance?)

Q4. How do you handle bikes that are returned by a customer and need refurbishment for another customer?

Q5. How long do bikes typically stay in each state?

Service agreements

Your whiteboard mentioned a key rule: "if service agreement = no invoice." We've modeled service agreements as customer-level contracts that determine whether maintenance work gets billed.

The system's view today: a service agreement covers a customer (not specific bikes), has a start and end date, and has flags for what's covered — parts, labor, or both. There's an optional monthly fee. When a maintenance work order is created for a bike whose customer has an active service agreement, the system automatically marks the parts and/or labor as covered, and the work order is closed without invoicing.

QUESTIONS FOR DENNIS — Service agreements

Q1. Is a service agreement always per-customer, or is it sometimes per-bike (e.g., "this fleet of 50 is covered, those 10 are not")?

Q2. What does "covered" actually mean in your agreements? Parts only? Labor only? Both? Limited number of repairs per year?

Q3. Do you have agreements that exclude specific things — e.g., cover everything except battery replacement, or except theft/damage?

Q4. How are agreements priced — monthly fee? Annual fee? Per-bike fee? Bundled into the bike sale?

Q5. When an agreement expires, what happens — automatic renewal, or back to invoicing every repair?

Q6. Do customers without an agreement still get repair history tracked, or only those who pay for it?

Multi-currency and international customers

Today most of your business is in DKK, but the system was designed from the start to handle multiple currencies and international customers. Specifically:

- Every monetary value is stored together with its currency. There are no "naked" amounts.
- When you buy parts in USD from Eastek HK, the system records the FX rate at the moment of purchase. The cost basis in DKK is locked in then — a later rate change doesn't rewrite history.
- When you sell to a customer outside Denmark, you can invoice in their currency or in DKK — whichever they prefer.
- Different VAT rules can be applied to different lines of the same invoice. Standard 25% Danish moms for domestic; 0% reverse-charge for EU B2B; 0% export for non-EU.

QUESTIONS FOR DENNIS — International

Q1. Do you currently sell to customers outside Denmark? Which countries, if so?

Q2. Have you ever invoiced in EUR, GBP, or another non-DKK currency? How often?

Q3. Are EU reverse-charge sales something you handle today, or all your sales currently domestic?

Q4. Do you import parts from anywhere besides Eastek HK (and in what currencies)?

Q5. For international shipping, do you record carriers and tracking numbers today? Use Incoterms (FOB, DAP, DDP)?

Suppliers and purchasing

The system tracks each supplier, the parts they offer, the price they charge, lead times, and the full history of every purchase. Each purchase remembers four things that matter for cost basis:

4. What you paid (e.g., 5.73 USD per cable)
5. The currency it was paid in (USD)
6. The exchange rate at that moment (e.g., 6.3164 USD→DKK)
7. Your transport markup factor (e.g., 1.10 = 10% added for shipping)

These four things together produce the landed cost in DKK per unit (5.73 × 6.3164 × 1.10 = 39.81 DKK in this example), which is what gets used for inventory valuation.

QUESTIONS FOR DENNIS — Suppliers

Q1. How many suppliers do you typically work with? (1–3? 5–10? More?)

Q2. Apart from Eastek HK, who are your other key suppliers?

Q3. Do prices fluctuate often, or are they fairly stable per supplier?

Q4. Is the 1.10 transport factor a constant for all your purchases, or does it vary by shipment?

Q5. When you receive a partial shipment (e.g., ordered 100, got 80), how do you want to handle that?

Q6. Do you have a preferred-supplier-per-part rule, or do you shop around?

Maintenance flow

When a customer has a problem with a bike, here's the flow:

8. A maintenance ticket is created — it captures who reported the issue, how (email, phone, app, in person), the bike's frame number, a description, and a priority level.
9. The ticket gets diagnosed. If parts are needed, status moves to "awaiting parts" and the right parts are pulled or ordered.
10. A work order is created — this is the actual job: assigned mechanic, time spent, parts used, work performed.
11. Each part used in the work order draws down inventory automatically.
12. When the work is done, the system checks: does this customer have an active service agreement? If yes, parts and/or labor are marked covered. If no, an invoice is generated.
13. The bike's status returns to "in service" and the work order is closed.

QUESTIONS FOR DENNIS — Maintenance

Q1. How do customers actually report problems today — phone, email, walk-in, an app, all of these?

Q2. Do you ever do preventive maintenance on a schedule (e.g., every 6 months for hospital fleets), or only reactive repairs?

Q3. When a bike comes in for repair, does it physically come back to your shop, or do you go to it on-site?

Q4. Who decides if a repair is covered by service agreement — you, the mechanic, automatic by rule?

Q5. Do you ever quote a customer for a repair before starting it (i.e., a maintenance offer)?

Q6. How do you handle warranty claims — different from service agreement?

Invoicing and accounting

The system can generate invoices automatically from sales orders, manufacturing-order completions, and billable work orders. Invoice numbers are issued as INV-2026-0001, INV-2026-0002, etc. — a clean year-prefixed sequence.

For Danish public-sector customers, invoices automatically include the EAN number on file. For EU B2B customers with valid VAT numbers, lines can be marked reverse-charge (0% VAT). For exports outside the EU, lines are zero-rated.

The system is designed to integrate with e-conomic via API — invoices issued in the FMS can be pushed to e-conomic for the actual accounting/bookkeeping. The FMS doesn't try to BE the accounting system; it produces the invoices and hands them off.

QUESTIONS FOR DENNIS — Invoicing

Q1. Do you use e-conomic today, or another accounting system (Billy, Dinero, Visma)?

Q2. How are invoices generated currently — in your accounting tool, in Word, on paper?

Q3. How often do you issue EAN-based public-sector invoices? Is that a major part of your business?

Q4. When a customer pays, do you want the system to track that, or is that purely the accounting tool's job?

Q5. Do you ever issue credit notes (kreditnota) for returns or corrections? How often?

Q6. What's your typical payment terms — 14 days net? 30 days?

What we're deliberately leaving for later

Your whiteboard included some great ideas that aren't in this version of the system — not because they're bad, but because they're each their own project. Here's what we're deferring, and roughly when we'd revisit them:

Feature	Why later
AI agents to respond to customer chats	Belongs after the core data is solid. Once the FMS knows your parts, customers, and bikes, an AI agent can be built on top.
Live streaming of bike builds	A marketing feature, not an operational one. Doesn't depend on the FMS.
Bike designer with prices on the homepage	Customer-facing feature; needs the parts catalog stable first, then a public website.
Camera control / automatic picture upload	Once you have the system running, integrating cameras is a small project.
Own telephone system	Big project unto itself; e-conomic integration and email integration are easier wins first.
AirTag / GPS / RFID tracking	The data model already supports these identifiers. Wiring them to live tracking comes when AirTags actually start being installed.
Preventive maintenance schedules	A natural extension to the maintenance module; comes after you're running reactive repairs in the system.

None of these are abandoned. The data structure has been designed so they can all be added later without disruption.

Sign-off summary

Here’s a one-page summary of the major decisions in the system that need your sign-off. Where you agree, just write "OK." Where you don’t, mark up what’s wrong.

Decision	What we’ve set
Bike types	Pedal, E-bike, Tricycle, Cargo, Hand-cycle, Child, Other
5 e-bike identifiers	Frame, Lock, Battery Lock, Battery, Charger
Pedal bike identifiers	Frame + Lock only
Customer segments	Hotel, Hospital, Municipality, FM, B2B, B2C, Rental, Other
Pricing groups	Retail 0% / Wholesale 15% / Corporate 20%
Bike lifecycle	Planning → Building → In Stock → Assigned → In Service → In Maintenance → Retired
Sales order vs Manufacturing order	Kept separate — sales = customer-facing, MO = internal build
Service agreement scope	Per-customer, with parts/labor coverage flags
Multi-currency support	All currencies; FX rate frozen at moment of purchase
VAT handling	Per-line; 25% domestic, 0% EU reverse charge, 0% non-EU export
Document numbering	Year-prefixed: INV-2026-0001, MO-2026-0001, etc.
Accounting integration	Push to e-conomic via API
GDPR compliance	Customer PII redactable on request; financial records retained 5 years

What happens after you sign off

Once you've given us your feedback, we will:

14. Fold your corrections into Schema v1.2.
15. Apply v1.2 to a real Supabase database hosted in the EU (for GDPR).
16. Begin building Phase 1 — the Parts catalog UI — in real code.
17. Show you working features as they're completed, not in a giant reveal at the end.

Phase 1 will let you operate parts inventory in the real system and retire the existing Excel. Subsequent phases bring online the Build Workbench, Customer/Fleet management, Maintenance, and Invoicing in that order.

Total estimated time to a fully operational core: roughly five months for one developer working with AI tools, faster with a second pair of hands.

Thank you for taking the time to review this.
The whole system depends on us getting your business right.